Benchmarking Counterfactual Explanations of Model Change with Exact Ground Truth

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Abstract. Counterfactual methods are used to explain how a classifier changed after an update, but without ground truth one cannot tell which method sees which kind of change. We contribute a benchmark with exact ground truth (rendered images, injected changes, a grid over the rotation and offset of the decision boundary) and a linear analysis of what counterfactual differencing, gradient differences, importance differences, disagreement search and differential counterfactuals (Δ-CE, minimal inputs on which exactly one model changes its decision by a margin) can see. The analysis predicts that gradient differences see a rotation and are blind to an offset; both halves were confirmed on CNNs in tests registered before the grid was run (Recovery 0.08 on offsets; +0.57 from the smallest to the largest rotation), and with quantities taken at the query it also predicts which witnesses show the new dependence (descriptive). A pre-registered hypothesis that Δ-CE recovers the changed factor better than gradient differences was rejected. Margin witnesses abstain on null updates, also on a tabular dataset. Thresholded to flag none of these nulls, a differencing score flags no query of constant offsets below $|b| = 2$ or of fine-tuned offsets (at most 0.28 of the latter at 5% false positives), while witnesses detect constant logit offsets from $|b| = 1.1$ and fine-tuned offsets on every lowered-class query in 6 of 7 pairs (descriptive).

1 Introduction

Deployed classifiers are updated routinely, and an update that improves accuracy can still change individual decisions in unwanted ways [11]; explaining *how* they differ is a goal of explainable AI under distributional change [3, 12]. Artelt et al. [1, 2] compare a counterfactual [8] of the same input under each model, or approximate this by a difference of gradients; Muschalik et al. [4] compare feature importances; disagreement search [5, 6, 7] looks for inputs on which two models disagree. These families assume different things about *what* changed, and without ground truth this goes unchecked.

We ask which kind of change each family can see. We also include *differential counterfactuals* (Δ-CE): the closest input on the data manifold at which only f_{new} changes its decision while f_{old} keeps it, both by a margin κ. Δ-CE anchors disagreement search [5, 6] at a query; unlike robust counterfactuals [9, 10], which seek a point valid for both models, it seeks one valid for exactly one, and the margin makes it return nothing for models that differ only by jitter.

Contributions. (i) A benchmark with exact ground truth for explaining model updates: five rendered factors, three injected changes, null updates, and a 4×4 grid over the rotation and threshold of the decision boundary (32 of the 49 model pairs), with code and protocol. (ii) A linear analysis separating rotation from offset, with a prediction for each family. (iii) A test of these predictions on

CNNs under a protocol fixed in advance. The analysis transfers with quantities taken at the query, and its projection equivalence only with the old normal taken at the witness; Table 3 separates tests that follow from a lemma or definition from risky predictions, and all failures are reported.

2 Differential counterfactuals and what each family sees

Let $D : \mathcal{Z} \to \mathcal{X}$ map a search space onto the data manifold (here an exact renderer from factor space $\mathcal{Z} = [0,1]^5$). For class y the margin is $m(z) = f_y(D(z)) - f_{1-y}(D(z))$; m_o, m_n belong to f_{old}, f_{new}, and $h = m_n - m_o$. A query z_0 is valid if $m_o(z_0) \geq \kappa$ and $m_n(z_0) \geq \kappa$.

Definition 1. With $\mathcal{F}_\kappa^N = \{z : m_n(z) \leq -\kappa,\ m_o(z) \geq \kappa\}$, a Δ-CE of z_0 is $z^\star \in \arg\min_{z \in \mathcal{F}_\kappa^N} \|z - z_0\|$ if $\|z^\star - z_0\| \leq r$, and NOT_FOUND otherwise; swapping the models gives type O. Each $z \in \mathcal{F}_\kappa^N$ is a *witness*: two forward passes verify that the models decide differently, both decisively, which a difference of two counterfactuals does not.

Lemma 1. Every witness has $h(z) \leq -\kappa - m_o(z) \leq -2\kappa$. Hence models whose margins differ by less than 2κ everywhere have no witness.

Linear case. For linear margins on $\mathbb{R}^p$ with independent w_o, w_n the nearest witness is unique: the ordinary counterfactual of f_{new} or the minimum-norm point with both constraints active. Write $\hat{w}_n = \cos\theta\, \hat{w}_o + \sin\theta\, v$ with $v \perp \hat{w}_o$, and let d_n, d_o be the distances from z_0 to $\{m_n = -\kappa\}$ and $\{m_o = \kappa\}$. The gradient difference of [1] is $-(\hat{w}_n - \hat{w}_o)$, counterfactual differencing $u_n - u_o$ (the two minimum-norm counterfactuals), the witness $u^\star$.

Proposition 2. (a) *Offset* ($\theta = 0$): the gradient difference is 0; counterfactual differencing is a multiple of $\hat{w}_o$; a witness exists iff $d_n \leq d_o$ and equals $-d_n \hat{w}_o$. (b) *Rotation* ($0 < \theta < \pi/2$): the gradient difference has components in ratio $\tan(\theta/2) < 1$ along $(\hat{w}_o, v)$, i.e. mainly along the new dependence v; the witness has $|u^\star \cdot \hat{w}_o| = \min\{d_n \cos\theta, d_o\}$ and is dominated by v iff $\theta > \pi/4$ or $d_o(\sin\theta + \cos\theta) < d_n$. After projecting out $\hat{w}_o$ all three coincide up to scale (cf. Theorem 1 of [1], which gives the cosine of two weight vectors from their counterfactuals).

On CNNs the last statement holds approximately only with $\hat{w}_o$ taken at the witness (Section 4.2). In words, gradient differences see a *rotation* and are blind to an *offset*; type-N witnesses exist only on the side whose margin the offset lowers; for a constant logit offset such as the prior correction [13], they abstain exactly when the shift is below 2κ (Corollary 1 in the proofs). The analysis was formulated after the main experiment and before the grid was run.

3 Benchmark and protocol

Data and models. Images are 32×32 ellipses rendered from five independent factors in $[0,1]$: aspect (task factor), size, colour, background and horizontal position; labels are $y = \mathbb{1}[\text{aspect} > 0.5]$ with 5% noise; the renderer is exact, so the factor change of any explanation is read without a learned oracle. f_{old} is a small CNN (five seeds; two on the grid); f_{new} is f_{old} fine-tuned on 10 000

Table 1: Injected changes; $k^\star$ is the ground-truth factor (none for the grid cell $(0, 0.5)$). [a]Two pairs have no $k^\star$ ($t = 0.5$) and two failed the offset check ($t = 0.53$), so four enter the offset test of Table 3.

Change	Adaptation data	$k^\star$	Pairs
Shortcut	colour correlated with label (0.99)	colour	3
Subgroup	labels flipped for size > 0.75	size	3
Threshold	$y = \mathbb{1}[\text{aspect} > 0.6]$	aspect	3
Null	resample / size $\sim U[0.5, 1]$, same rule	—	5 / 3
Grid, $\alpha = 0$	$y = \mathbb{1}[\text{aspect} + \alpha(\text{colour} - 0.5) > t]$, $t \in \{.5, .53, .56, .6\}$	aspect	8[a]
Grid, $\alpha > 0$	same, $\alpha \in \{.1, .25, .5\}$	colour	24

samples from one condition (Table 1). A pair enters only if its change is visible in decisions on validation data (null pairs: decisive disagreement ≤ 0.02). For the grid we measure the realised rotation $\hat{\alpha}$ and threshold $\hat{t}$ of each f_{new} by a logistic fit to its decisions, and analyse at these measured values; all 32 realised the intended rotation, while offsets below about 0.05 were not produced reliably.

Metrics and methods. Per pair: $N = 300$ valid test points. Recovery@r is the share of points with a valid explanation within the budget whose largest factor change is $k^\star$ (NOT_FOUND and ties count 0). Recovery uses type-N witnesses only (pre-declared); a query whose class the update raises (Threshold, class 0) has none and counts 0. All counterfactual methods share one solver (Adam, then SLSQP), the points and the budget $r = 0.54$, set once on validation. We compare Δ-CE ($\kappa = 0.5$) with **CF-diff**, counterfactual differencing [1]; **Grad-diff**, the gradient difference [1]; **Imp-diff**, a permutation-importance difference [4]; **Seg-search**, a line search along segments to training points, after [5]; **No-prox**, Δ-CE without the proximity term (the DeepXplore objective [6]); and **CF-new**, the ordinary counterfactual of f_{new} alone.

Protocol. Every test split served at most one confirmatory test, after all choices were fixed: split A is the locked split of Table 2, split B the new data of the tests formed after the first results, split C the fresh data of the type-O analysis. Table 3 lists every test; the grid's manipulation check, exclusion rule and statistics were fixed after a pilot and before its test data; one analysis-code error was later corrected. Bootstraps are paired and resample points within pairs (three per condition; two seeds of f_{old} on the grid), so intervals do not cover model-to-model variation. The rest is descriptive. A digit tier and a label-shift update failed qualification.

4 Results

4.1 Offsets

Gradient differences are blind to an offset, as Proposition 2(a) predicts: on the pure offsets of the grid their Recovery is 0.081 (below 0.2 in each of the 4 pairs), and on Threshold Δ-CE beats them (0.41 vs. 0.10 on split B; positive for each of the three seeds). On an offset the changed factor is the task factor, so even

Table 2: Recovery@r (found-rate) with type-N Δ-CE, split A (locked), mean over three pairs (95% CIs within ± 0.035); last row: found-rate on the 8 null pairs.

	Δ-CE	CF-diff	Grad-diff	Imp-diff	Seg-search	No-prox	CF-new
Shortcut	0.38 (0.84)	0.54 (0.93)	**0.91** (1.0)	0.00	0.02 (0.08)	0.35 (0.63)	0.07 (0.93)
Subgroup	0.53 (0.81)	0.32 (0.99)	**0.62** (1.0)	0.00	0.06 (0.10)	0.29 (0.58)	0.30 (1.0)
Threshold	0.47 (0.47)	0.76 (0.90)	0.07 (1.0)	**1.00**	0.07 (0.08)	0.31 (0.40)	0.90 (0.90)
Null	**0.000**	0.994	1.0	1.0	0.000	0.000	0.996

Table 3: Tests, grouped as in Section 4 (95% CIs; †90%). K: D = follows from a lemma, proposition or definition; R = risky prediction. When formed: pre = before any result; post = after the first, on split B; grid = before the grid run; new = later, new data. Holm families: *, ‡, §. Proj.: $\hat{w}_o$ projected out. Protocol IDs: repository.

Claim	Data	Estimate	p_{Holm}	Status	K	When
Offsets						
Grad-diff < 0.2	Grid, $\alpha = 0$	0.081 [0.066, 0.096]	$4{\cdot}10^{-4}$ ‡	confirmed	R	grid
Grad-diff offset-free	Grid	-0.098 [-0.143, -0.053]†	0.94 ‡	not conf.	R	grid
Δ-CE > Grad-diff	Threshold, B	$+0.31$ [0.27, 0.35]	$2{\cdot}10^{-4}$ *	confirmed	D	post
Rotations						
Grad-diff rises with α	Grid	$+0.57$ [0.53, 0.61]	$4{\cdot}10^{-4}$ ‡	confirmed	R	grid
Proj. Δ-CE $\approx$ Grad-diff	Shortcut, B	-0.30 [-0.33, -0.28]†	1 *	not conf.	R	post
Proj. Δ-CE $\approx$ Grad-diff	Grid	-0.35 [-0.38, -0.33]†	1 ‡	not conf.	R	grid
Normal at $z^{\star}$ halves gap or more	Grid, new	$-0.19 \to +0.03$ [0.01, 0.05]†	$6{\cdot}10^{-4}$ §	confirmed	R	new
Abstention						
No witness on nulls	Null, B; Adult	0/2 400; 0/1 500		confirmed	D	post; new
CF-diff (no cut) flags $\lvert b\rvert < 2\kappa$	Logit offsets	0.961 [0.955, 0.967]	$2{\cdot}10^{-4}$ §	confirmed	D	new
Δ-*CE versus gradient differences*						
Δ-CE > Grad-diff	Subgroup	-0.09 [-0.13, -0.05]		rejected	R	pre
Tie ($\lvert\Delta\rvert \leq 0.1$)	Shortc., Thresh.	$\lvert\Delta\rvert = 0.17$–$0.53$		not conf.	R	pre

CF-new recovers it (0.90–0.98). Type-N witnesses exist only on the side the offset lowers (on Threshold: every class-1 query, none of class 0), which is why Δ-CE scores 0.47 in Table 2; type O covers the other side. On split C (three Threshold pairs, 900 queries, descriptive) type N finds a witness for every class-1 query and none of class 0 (Recovery 0.45), type O the converse, and the nearer of the two recovers aspect on all 900 queries (1.00; CF-diff 0.77, CF-new 0.89, Grad-diff 0.07). For an exact logit offset b, witnesses appear only once $|b|$ reaches 2κ and then, empirically, on every query of the lowered class. CF-diff, with no threshold, flags 0.961 of queries already at $|b| \in \{0.5, 0.9\}$, where 3–8% of decisions do flip; the witness misses this real change by design (Corollary 1), a difference in sensitivity, not an error of CF-diff.

4.2 Rotations

Gradient differences see a rotation, increasingly with its angle: Grad-diff Recovery rises from 0.36 at $\alpha = 0.1$ to 0.93 at $\alpha = 0.5$ (each $\alpha = 0.5$ pair above each $\alpha = 0.1$ pair; Fig. 1), while CF-new recovers colour on ≤ 0.03 of points. Proposition 2(b) predicts that after projecting out $\hat{w}_o$ witnesses and gradient differences coincide up to scale; with $\hat{w}_o$ taken at the query they do not (Shortcut 0.64 vs. 0.95 on

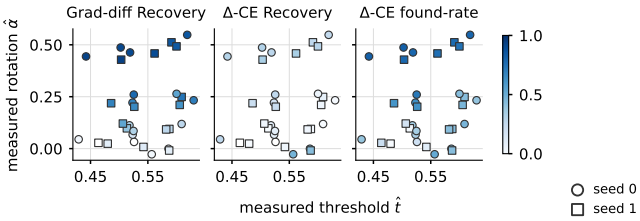


Fig. 1: Grid results at the measured rotation $\hat{\alpha}$ and threshold $\hat{t}$ of each f_{new} (grid test split), on one colour scale from 0 to 1. Hollow: no $k^\star$; marker shape: seed of f_{old}.

split B, all valid points; grid -0.35). A post-hoc explanation, tested on new data (four grid pairs, points with a witness): with $\hat{w}_o$ taken at the witness the projected gap Δ-CE − Grad-diff moves from -0.190 to $+0.029$ (narrowing by more than half in each of the 4 pairs, by a small margin in two). This locates the failure, not its cause. On Shortcut (points with a witness) it moves from -0.166 to -0.024; the normal turns by a median 32° from z_0 to $z^\star$, not more for distant witnesses as registered. The witness direction itself follows the analysis: where the condition of Proposition 2(b), computed from margins and gradients at z_0, holds, witnesses recover colour on 0.64 of points, elsewhere on 0.04 (24 rotation pairs, new data, 3 952 witnesses; gap positive in each of the 21 pairs with both groups; descriptive).

4.3 Null updates and abstention

All three disagreement searches verify the margin, and none returns a point on a null update (Table 2). Δ-CE adds proximity: No-prox finds witnesses on fewer points (0.40–0.63 vs. 0.47–0.84) and, where both find one, farther away on every point (median 0.31–0.33 vs. 0.20–0.23). On the 8 null pairs and on five null updates of an MLP on Adult (1 500 test points) no witness exists, while CF-diff attributes a change to 97.7–100% of points. Part of this contrast is the threshold. Scoring a query by $\|u_n - u_o\|$ and cutting on the 2 400 null queries so that none is flagged, as for Δ-CE, CF-diff flags at most 0.001 of the logit-offset queries up to $|b| = 2$, while Δ-CE has a witness for every lowered-class query from $|b| = 1.1$. For a constant offset Corollary 1 makes the witness an exact test of $|b| \geq 2\kappa$, so we repeated this on fine-tuned offsets, where it is not (three Threshold pairs of split C, four $\alpha = 0$ grid pairs): at the same cut CF-diff flags no lowered-class query in any pair, and only 0.05–0.28 at 5% false positives, while Δ-CE has a witness for every one in 6 of 7 pairs and 0.25 in the seventh (descriptive, saved outputs). Null pairs were admitted only without decisive disagreement on validation, which is close to the witness criterion, so the 5% point is the fairer reading. NOT_FOUND is no solver failure: 10^5 samples per query find none on 550 null or impossible-class queries.

5 Discussion and limitations

Witnesses are meant to show *where* decisions changed near a query. We have partial checks, not a score: each witness is verified (by definition), the search misses few (in the sampling audit, 3 of 395 small-offset queries) and returns nearer witnesses than No-prox; other families return directions, so this is no comparison. **Limitations.** (1) The image data are synthetic; the tabular test covers abstention only; of label shift, only the prior correction [13] was tested. (2) κ is an exact threshold only for a constant offset; otherwise abstention tracks decisive change (at shifts of 0.03–0.04 it depended on the fine-tuning recipe). (3) Witnesses see only decision changes on the manifold within the budget, with no certificate for deep networks. (4) Recovery scores one notion of change, the factor along which the weights moved. (5) Several tests were formed after earlier results, one after a validation run showed the effect. Every protocol change was logged, with dated commits, before the data it concerns were generated (the analysis-code correction excepted); log, proofs, code: [repository link].

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